NATIONAL TEACHERS COLLEGE
School of Arts, Sciences, and Technology
629 J. Nepomuceno St., Quiapo, Manila

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Date: October 23, 2026
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Week 10: Object-Oriented Programming in Dart (Part 2)
Your Turn: Submit Your Enhanced OOP App

Output:

```
=== ANIMAL MANAGEMENT SYSTEM ===

--- Animal Information ---
Name: Cali
Age: 3 years old
Cali says: Woof! Woof!

--- Animal Information ---
Name: Luna
Age: 2 years old
Luna says: Meow! Meow!

--- Animal Care Volunteer ---
Volunteer: Bhea
Training activity: Basic animal care

Program completed successfully.
```

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Code:

DartPad Create Create with Gemini Samples Run

```dart
abstract class Animal {
  final String name;
  final int age;

  Animal(this.name, this.age);

  // Abstract method that subclasses must implement.
  void makeSound();

  // Shared method inherited by subclasses.
  void showDetails() {
    print('Name: $name');
    print('Age: $age years old');
  }
}

class Dog extends Animal {
  Dog(String name, int age) : super(name, age);

  @override
  void makeSound() {
    print('$name says: Woof! Woof!');
  }
}
```

Run

```dart
}

class Cat extends Animal {
  Cat(String name, int age) : super(name, age);

  @override
  void makeSound() {
    print('$name says: Meow! Meow!');
  }
}

// Mixin for reusable training behavior.
mixin Trainable {
  void train(String skill) {
    print('Training activity: $skill');
  }
}

// This class uses the mixin without extending Animal.
class AnimalCareVolunteer with Trainable {
  final String name;

  AnimalCareVolunteer(this.name);
```

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```
  void introduce() {
    print('Volunteer: $name');
  }
}

void main() {
  print('=== ANIMAL MANAGEMENT SYSTEM ===\n');

  // Polymorphism: one list contains different Animal subclasses.
  List<Animal> animals = [
    Dog('Cali', 3),
    Cat('Luna', 2),
  ];

  for (Animal animal in animals) {
    print('--- Animal Information ---');
    animal.showDetails();
    animal.makeSound();
    print('');
  }

  // Demonstrate the mixin using the volunteer.
  print('--- Animal Care Volunteer ---');
```

```
    Dog('Cali', 3),
    Cat('Luna', 2),
  ];

  for (Animal animal in animals) {
    print('--- Animal Information ---');
    animal.showDetails();
    animal.makeSound();
    print('');
  }

  // Demonstrate the mixin using the volunteer.
  print('--- Animal Care Volunteer ---');

  AnimalCareVolunteer volunteer =
      AnimalCareVolunteer('Bhea');

  volunteer.introduce();
  volunteer.train('Basic animal care');

  print('\nProgram completed successfully.');
}
```